

Scale invariance beyond self-similarity

Danijel Grahovac

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School of Applied Mathematics and Informatics, J. J. Strossmayer University of Osijek, Croatia

Motivation

Motivation – Taylor's law

- Suppose we measure a numerical quantity across N groups:

$$X_i^{(j)}, \quad i = 1, \dots, n_j, \quad j = 1, \dots, N$$

- Compute the sample mean and variance for each group $j = 1, \dots, N$

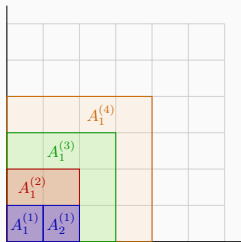
$$M^{(j)} = \frac{1}{n_j} \sum_{i=1}^{n_j} X_i^{(j)}, \quad V^{(j)} = \frac{1}{n_j - 1} \sum_{i=1}^{n_j} (X_i^{(j)} - M^{(j)})^2.$$

- Taylor's law:** for some $a, b \in \mathbb{R}$

$$V^{(j)} \approx a \left(M^{(j)} \right)^b, \quad j = 1, \dots, N,$$

- $b \in (0, 2]$ typically
- Thousands of papers after Taylor (1961)
- Many explanations:
 - power variance function family of distributions Kendal (2002)
 - population models Cohen, Xu & Schuster (2013)
 - infinite mean/variance Brown, Cohen & de la Peña (2017), Cohen, Davis & Samorodnitsky (2020), De La Pena, Doukhan & Salhi (2022), Cohen, Davis & Samorodnitsky (2022)

Example. Quadrat count of Colorado potato beetle Taylor (1961), Beall (1938)



Model:

- $X_i^{(j)} = \xi(A_i^{(j)})$ for some random measure ξ
- Assume that $A_i^{(j)} = c_i^{(j)} + \lambda_j B$, $i = 1, \dots, n_j$
- If ξ is stationary, (population) Taylor's law may be stated as:
$$\text{Var } \xi(\lambda_j B) = a(\mathbb{E} \xi(\lambda_j B))^b, \quad j = 1, \dots, N$$

Taylor's law formulation

Population Taylor's law: for some $a, b \in \mathbb{R}$

$$\text{Var } \xi(\lambda B) = a (\mathbb{E} \xi(\lambda B))^b, \quad \lambda \in (0, 1].$$

(or asymptotically for $\lambda \rightarrow 0$)

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Generalized population Taylor's law: for some $a, b: (0, \infty) \rightarrow \mathbb{R}$

$$\mathbb{E} \xi(\lambda B)^q = a(q) (\mathbb{E} \xi(\lambda B))^{b(q)}, \quad \lambda \in (0, 1],$$

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What kind of random measures have this property?

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What kind of random measures have this property?

- If ξ is a stationary completely random measure, then $b = 1$ (e.g. Poisson)
- If ξ is self-similar ($\xi(\lambda \cdot)$ and $\lambda^H \xi(\cdot)$ have the same distribution), then $\mathbb{E} \xi(\lambda B) = \lambda^H \mathbb{E} \xi(B)$, $\text{Var } \xi(\lambda B) = \lambda^{2H} \text{Var } \xi(B)$ and $b = 2$

How to explain other values of b ?

Temporal Taylor's law: for stochastic process $X = \{X(t)\}$

$$\text{Var } X(\lambda t) = a (\mathbb{E}X(\lambda t))^b, \quad \lambda \in (0, 1].$$

or

$$\mathbb{E}X(\lambda t)^q = a(q) (\mathbb{E}X(\lambda t))^{b(q)}, \quad \lambda \in (0, 1]$$

- If X is a subordinator, $b = 1$
 - If X is a nonnegative self-similar process, $b = 2$
-
- ▶ For other values of b we need more general scale invariance property
 - ▶ What is possible with integer-valued processes?

Stochastic self-similarity (multifractality)

Typical properties considered as **multifractality** (**stochastic self-similarity** is probably a better term):

Exact scaling of moments:

$$\mathbb{E}|X(t)|^q = c(q)t^{\tau(q)}, \quad (1)$$

where the **scaling function** $q \mapsto \tau(q)$ is nonlinear (for H -self-similar $\tau(q) = Hq$)

Asymptotic scaling of moments: (1) holds as $t \rightarrow 0$

Exact stochastic scale invariance (statistical/stochastic self-similarity): for every $\lambda \in \Lambda \subset (0, \infty)$ there exists a positive random variable M_λ independent from the process $\{X(t)\}$ such that

$$X(\lambda t) \stackrel{d}{=} M_\lambda X(t)$$

or

$$\{X(\lambda t)\} \stackrel{fdd}{=} \{M_\lambda X(t)\}$$

The main example

Multiplicative cascade processes

- Introduced by Mandelbrot (1972) using a discrete grid-based construction and later extended by Barral & Mandelbrot (2002), Bacry & Muzy (2003), Muzy & Bacry (2002) and Barral & Jin (2014)
- Cascade process $\{X(t)\}$ has the stochastic scale invariance property – for $\lambda \in (0, 1]$

$$\{X(\lambda t)\}_{t \in [0, T]} \stackrel{fdd}{=} \{W_\lambda X(t)\}_{t \in [0, T]},$$

where

- $W_\lambda = \lambda e^{\Omega(\lambda)}$ is independent of $\{X(t)\}$
- $\mathbb{E}[e^{\theta \Omega(\lambda)}] = \lambda^{-\psi(\theta)}$ with ψ the Laplace exponent of some infinitely divisible distribution
- T is the *integral scale*
- In particular, exact scaling of moments holds with $\tau(q) = q - \psi(q)$ since

$$\mathbb{E}|X(t)|^q = \mathbb{E}\left[t^q e^{q\Omega(t)}\right] \mathbb{E}|X(1)|^q = t^{q-\psi(q)} \mathbb{E}|X(1)|^q, \quad t \in (0, 1].$$

Other examples

- Combining a cascade process with a self-similar process

Goal: define and study multifractal processes as a generalization of self-similar processes

- The property

$$\{X(\lambda t)\}_{t \in \mathcal{T}} \stackrel{fdd}{=} \{M_\lambda X(t)\}_{t \in \mathcal{T}}$$

requires for $\lambda \in \Lambda$ the existence of a *single* random variable M_λ such that

$$(X(\lambda t_1), \dots, X(\lambda t_n)) \stackrel{d}{=} (M_\lambda X(t_1), \dots, M_\lambda X(t_n))$$

for any choice of $t_1, \dots, t_n \in \mathcal{T}$ and $n \in \mathbb{N}$

- Instead we will require only that the random factors corresponding to each time point are identically distributed

Definition

A stochastic process $X = \{X(t), t \in \mathcal{T}\}$ is **multifractal** if there exist sets $\Lambda \subset (0, \infty)$ and $\mathcal{S} \subset \mathcal{T}$ such that for every $\lambda \in \Lambda$ there exists a family of identically distributed positive random variables $\{M_\lambda(t), t \in \mathcal{S}\}$, independent of $\{X(t)\}$ such that

$$\{X(\lambda t)\}_{t \in \mathcal{S}} \stackrel{fdd}{=} \{M_\lambda(t)X(t)\}_{t \in \mathcal{S}}. \quad (2)$$

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- (2) means that for every choice of $t_1, \dots, t_n \in \mathcal{S}$, $n \in \mathbb{N}$ it holds that

$$(X(\lambda t_1), \dots, X(\lambda t_n)) \stackrel{d}{=} (M_\lambda(t_1)X(t_1), \dots, M_\lambda(t_n)X(t_n)).$$

- Instead of $M_\lambda(t) = M_\lambda$ we only assume that

$$M_\lambda(t_1) \stackrel{d}{=} M_\lambda(t_2), \quad \forall t_1, t_2 \in \mathcal{S}.$$

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There are three sets:

- $\mathcal{T}$ is the time set where the process is defined (we assume $\mathcal{T} = (0, \infty)$, $\mathcal{T} = [0, \infty)$ or $\mathcal{T} = (0, T]$, $\mathcal{T} = [0, T]$ for some $T > 0$)

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- Λ is a set of scales for which (2) holds
- $\mathcal{S}$ is a subset of $\mathcal{T}$ over which the equality of finite dimensional distributions (2) holds

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- **Example.** Suppose $\Lambda = (0, \infty)$, $\mathcal{S} = \mathcal{T}$ and for every $\lambda \in \Lambda$, $M_\lambda(t)$ is deterministic. Then $t \mapsto M_\lambda(t)$ is constant, say m_λ , and $\{X(t)\}$ is self-similar.

- If the scaling is deterministic, then typically $\Lambda = (0, \infty)$ and $\mathcal{S} = \mathcal{T}$.

Proposition

Suppose X is a multifractal process such that $M_\lambda(t) = m_\lambda$ is deterministic for every $\lambda \in \Lambda$.

- *If Λ contains an interval, then (2) holds for any $\lambda \in (0, \infty)$.*
- *If additionally $\mathcal{S}$ contains an interval, then (2) holds with $\mathcal{S} = \mathcal{T}$, hence X is self-similar.*

$$\{X(\lambda t)\}_{t \in \mathcal{S}} \stackrel{fold}{=} \{M_\lambda(t)X(t)\}_{t \in \mathcal{S}}. \quad (2)$$

- A kind of converse holds: scaling with the random scaling factor cannot hold for any scale since then it reduces to deterministic scaling

Proposition

Suppose X is a multifractal process with $\bar{q}(X) = \sup \{q \geq 0 : \mathbb{E}|X(t)|^q < \infty, t \in \mathcal{T}\} > 0$. If $\Lambda = (0, \infty)$, then $M_\lambda(t) = m_\lambda$ is a.s. a constant for any $\lambda \in \Lambda'$, $\Lambda' = \{\lambda \in \Lambda : \text{there exists } t \in \mathcal{S} \text{ such that } \lambda t \in \mathcal{S}\}$. In particular, if $\mathcal{S} = \mathcal{T}$, then X is self-similar.

- $\Lambda' = (0, \infty)$ as soon as $(0, u) \subset \mathcal{S}$ or $(u, \infty) \subset \mathcal{S}$ for some $u > 0$
- We will have two typical classes of processes with stochastic scale invariance depending on whether $\Lambda = (0, 1]$ or $\Lambda = [1, \infty)$

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The set $\mathcal{S}$

- For random scaling one has to make restrictions on the set $\mathcal{S}$ too

Proposition

Suppose $\mathcal{T} = (0, \infty)$ or $\mathcal{T} = [0, \infty)$ and X is a jointly measurable multifractal process such that $\bar{q}(X) > 0$ and $\Lambda = (0, 1]$ or $\Lambda = [1, \infty)$. If

$$\{1/\lambda : \lambda \in \Lambda\} \subset \mathcal{S},$$

then X is self-similar.

- Random scaling is impossible if
 - $\Lambda = (0, 1]$ and $[1, \infty) \subset \mathcal{S}$, or
 - $\Lambda = [1, \infty)$ and $(0, 1] \subset \mathcal{S}$,or with $\mathcal{S} = (0, \infty)$
- $\mathcal{S}$ can be extended (e.g. by putting $\tilde{X}(t) = X(t/T)$)

Summary: $\Lambda = (0, 1]$, $\mathcal{S} = (0, 1]$ or $\mathcal{S} = (0, T]$, $\mathcal{T} = (0, \infty)$ or $\mathcal{T} = (0, T]$

Proposition

If X is multifractal and $\bar{q}(X) > 0$, then the following holds:

- (i) $M_1(t) = 1$ a.s. for every $t \in S$.*
- (ii) Let $\lambda \in \Lambda$ and $t \in S$. Then for every $n \in \mathbb{N}$ there exist independent identically distributed positive random variables $M^{(1)}, \dots, M^{(n)}$ such that*

$$M_\lambda(t) \stackrel{d}{=} M^{(1)} \dots M^{(n)}.$$

Moreover, $M^{(1)} \stackrel{d}{=} M_{\lambda^{1/n}}(t)$.

- (ii) is known as log-infinite divisibility (Veneziano (1999), Bacry & Muzy (2003)), multiplicative infinite divisibility Hirsch & Yor (2013) or M -infinite divisibility Zolotarev (1986)
- (ii) implies that for every $\lambda \in \Lambda$ and $t \in S$, $\log M_\lambda(t)$ is infinitely divisible

- Consider the process $\{\log M_{e^{-s}}(t), s \geq 0\}$

Proposition

Suppose X is multifractal, $\bar{q}(X) > 0$ and for some $t \in S$ (and hence for every $t \in S$), $\{M_\lambda(t), \lambda \in \Lambda\}$ is continuous in law.

Let $\{L(s), s \geq 0\}$ be a Lévy process such that

$$L(1) \stackrel{d}{=} \log M_{e^{-1}}(t).$$

Then for every $s \geq 0$

$$L(s) \stackrel{d}{=} \log M_{e^{-s}}(t).$$

- However, $\{\log M_{e^{-s}}(t), s \geq 0\}$ need not be a Lévy process

Proposition

Under the assumptions of previous proposition, for every $q \in \mathbb{R}$ such that $\mathbb{E}|X(t)|^q < \infty$ it holds that

$$\mathbb{E}|X(t)|^q = t^{-\psi(q)} \mathbb{E}|X(1)|^q, \quad t \in (0, 1],$$

where $\psi(q) = \log \mathbb{E}e^{qL(1)}$ is the Laplace exponent of the Lévy process L defined in previous proposition.

- The **scaling function** τ in scaling of moments is $-\psi$ which is strictly concave if $L(1)$ is non-degenerate
- **Taylor's law** – for X nonnegative we have

$$\mathbb{E}X(\lambda t)^q = a(q) (\mathbb{E}X(\lambda t))^{\frac{\psi(q)}{\psi(1)}}, \quad \lambda \in (0, 1],$$

$$\text{Var } X(\lambda t) \sim a(\mathbb{E}X(\lambda t))^{\frac{\psi(2)}{\psi(1)}}, \quad \lambda \rightarrow 0.$$

- The **cascade process** $\{X(t), t \in [0, \infty)\}$ satisfies for $\lambda \in (0, 1]$

$$\{X(\lambda t)\}_{t \in [0, T]} \stackrel{fdd}{=} \{W_\lambda X(t)\}_{t \in [0, T]},$$

- It is multifractal with $\mathcal{T} = [0, \infty)$, $\mathcal{S} = [0, T]$, $\Lambda = (0, 1]$
- The family of scaling factors $M_\lambda(t) = W_\lambda$ does not depend on t
- One-dimensional marginals of $\{\log W_{e^{-s}}, s \geq 0\}$ are the same as for some Lévy process, but $\{\log W_{e^{-s}}\}$ is not a Lévy process (fake Lévy process)

Idea:

- Construct a multifractal process so that $\{\log M_{e^{-s}}(t), s \geq 0\}$ is indeed a Lévy process

Lamperti transformation for self-similar processes:

- If $\{Y(t), t \in \mathbb{R}\}$ is a stationary process and $H \in \mathbb{R}$, then the process defined by

$$X(t) = t^H Y(\log t), \quad t > 0,$$

is H -self-similar

Theorem

Let $L = \{L(s), s \geq 0\}$ be a Lévy process and $Y = \{Y(u), u \in \mathbb{R}\}$ a stationary process independent of L . For $a \geq 0$ define the process $\{X^{(a)}(t), t \in (0, e^a]\}$ by setting

$$X^{(a)}(t) = e^{L(a - \log t) - L(a)} Y(\log t).$$

Then there exists a process $X = \{X(t), t > 0\}$ such that for every $a \geq 0$

$$\{X(t)\}_{t \in (0, e^a]} \stackrel{fdd}{=} \{X^{(a)}(t)\}_{t \in (0, e^a]}.$$

and for every $\lambda \in (0, 1]$

$$\{X(\lambda t)\}_{t > 0} \stackrel{fdd}{=} \{M_\lambda(t)X(t)\}_{t > 0},$$

where $\{M_\lambda(t), \lambda \in (0, 1], t > 0\}$ is the family of scaling factors corresponding to a Lévy process L .

The process X and the family $\{M_\lambda(t), \lambda \in (0, 1], t \in (0, 1]\}$ are independent, hence X is multifractal with $\mathcal{S} = \Lambda = (0, 1]$.

- For $L(s) = -Hs$, $s \geq 0$ one obtains the classical Lamperti transformation
- X typically does not have stationary increments

Scale invariance of integer-valued processes

Thinning operations

- $\mathbb{N}_0$ -valued processes cannot be self-similar ($X(\lambda t) \stackrel{d}{=} \lambda^H \cdot X(t)$)
- **van Harn & Steutel (1985)**: replace multiplication with thinning operation (*discrete multiplication*)
- Every natural thinning operation corresponds to a continuous composition semigroup of pgfs $F = (F_s)_{s \geq 0}$ such that **van Harn, Steutel & Vervaat (1982)**

$$F_t \neq I, t > 0 \text{ and } F'_1(1) = e^{-1}$$

- This leads to (**generalized**) **thinning operation** $\odot_F$:

$$P_{\alpha \odot_F Y}(z) = P_Y(F_{-\log \alpha}(z)), \quad \alpha \in (0, 1)$$

- **F-branching process** – a subcritical continuous-time Markov branching process $Z = \{Z(t), t \geq 0\}$ such that $Z(0) = 1$ a.s. and $P_{Z(s)}(z) = F_s(z)$
- For $Z_1, Z_2, \dots$ independent F -branching processes, independent of Y ,

$$\alpha \odot_F Y \stackrel{d}{=} Z_1(-\log \alpha) + \dots + Z_Y(-\log \alpha)$$

- $F_s(z) = 1 - e^{-s} + e^{-s}z$ corresponds to **binomial thinning** and pure-death process

Thinning for processes

- Let X be an $\mathbb{N}_0$ -valued stochastic process and $Z_1, Z_2, \dots$ are independent F -branching processes, independent of X
- For $\alpha \in (0, 1)$ we define the process $\{\alpha \odot_F X(t), t \geq 0\}$ by

$$\alpha \odot_F X(t) = Z_1(-\log \alpha) + \dots + Z_{X(t)}(-\log \alpha)$$

van Harn & Steutel (1985):

- An $\mathbb{N}_0$ -valued process X is (**discrete**) **H -self-similar**, $H \geq 0$, if for any $\lambda \in (0, 1]$

$$\{X(\lambda t)\} \stackrel{fdd}{=} \{\lambda^H \odot_F X(t)\}$$

- Let B denote the pgf $B(z) = \lim_{t \rightarrow \infty} \frac{F_t(z) - F_t(0)}{1 - F_t(0)}$, $0 \leq z \leq 1$
- X is self-similar if and only if

$$\{X(t)\} \stackrel{fdd}{=} \{N(Y(t))\},$$

where Y is an $\mathbb{R}_+$ -valued (classically) self-similar process and N a compound Poisson process with rate 1 and jump distribution with pgf B , independent of Y

- Thinning can be generalized random multipliers: for $(0, 1)$ -valued random variable A independent of X and $Z_1, Z_2, \dots$

$$A \odot_F X(t) = Z_1(-\log A) + \dots + Z_{X(t)}(-\log A)$$

- Consistent with other definitions and has natural properties,
e.g. $\{A \odot_F (B \odot_F X(t))\} \stackrel{d}{=} \{(AB) \odot_F X(t)\}$

Definition

An $\mathbb{N}_0$ -valued stochastic process X is (discrete) multifractal if for some $T > 0$ and every $\lambda \in (0, 1)$ there exists a $(0, 1)$ -valued random variable M_λ , independent of X , such that

$$\{X(\lambda t)\}_{t \in (0, T]} \stackrel{fdd}{=} \{M_\lambda \odot_F X(t)\}_{t \in (0, T]}.$$

Theorem

Let Y be an $\mathbb{R}_+$ -valued nondecreasing multifractal process $(\{X(\lambda t)\}_{t \in (0,1]} \stackrel{fdd}{=} \{M_\lambda X(t)\}_{t \in (0,1]})$ and N a compound Poisson process with rate 1 and jump distribution with pgf B , independent of Y . Then

$$X(t) = N(Y(t)), \quad t \geq 0,$$

is multifractal. For $n \in \mathbb{N}$ such that $\mathbb{E}Y(t)^n < \infty$ it holds that

$$\mathbb{E}(X(t))^n = \sum_{k=1}^n C_n(k) t^{\tau(k)}, \quad t \in (0,1],$$

where τ is the scaling function of Y and $C_n(k)$ are some constants.

- Factorial moments have simpler form

$$\mathbb{E}X(t)^{[n]} = \mathbb{E}X(t)(X(t) - 1) \cdots (X(t) - n + 1) = C_n t^{\tau(n)}$$

Scale invariance of random measures

- Self-similar random measures Vere-Jones (2005), Vervaat (1985) (there are other approaches, e.g. Hutchinson (1981), Zähle (1988) ...)
- Strictly α -stable random measures Davydov, Molchanov & Zuyev (2011)
- Discrete scale invariance: Mandelbrot's (canonical) multiplicative cascade
- $\star$ -scale invariance Rhodes, Sohier & Vargas (2014), Allez, Rhodes & Vargas (2013)
- Exact stochastic scale invariance: multifractal random measures (continuous cascades) introduced in Barral & Mandelbrot (2002), Bacry & Muzy (2003), Muzy & Bacry (2002) ($d = 1$), and Rhodes & Vargas (2010) ($d > 1$)

Scale invariance of point processes

- **Discrete stability** Davydov, Molchanov & Zuyev (2011): A point process Φ is discrete α -stable if for any $\lambda \in (0, 1]$

$$\lambda^{1/\alpha} \odot \Phi' + (1 - \lambda)^{1/\alpha} \odot \Phi'' \stackrel{d}{=} \Phi,$$

where Φ' and Φ'' are independent copies of Φ and $\alpha \odot \Phi$ is binomial thinning of point process with the p.g.fl.

$$G_{\alpha \odot \Phi}[u] = \mathbb{E} \exp \left\{ \int \log u(x) (\alpha \odot \Phi)(dx) \right\} = G_{\Phi}[1 - \alpha(1 - u)]$$

- **Branching-stable point processes** Zanella & Zuyev (2015): extending previous to generalized thinning involving branching process corresponding to semigroup F
- Another approach is to define branching-stability via scaling of point locations Bertoin, Cortines & Mallein (2018)

Self-similar point processes

Definition

A point process Φ on $\mathbb{R}^d$ is **self-similar** if for any $\lambda \in (0, 1]$

$$\Phi(\lambda \cdot) \stackrel{d}{=} \lambda^H \odot_F \Phi(\cdot).$$

- Let Z^∞ denote the random variable with pgf

$$B(z) = \lim_{t \rightarrow \infty} \frac{F_t(z) - F_t(0)}{1 - F_t(0)}, \quad 0 \leq z \leq 1,$$

and $(Z^\infty)_x$ denotes a point process having Z^∞ points in x and no points in $\mathbb{R}^d \setminus \{x\}$

Theorem

Let ξ be a self-similar random measure on $\mathbb{R}^d$ and define Π_ξ to be a Cox process with intensity measure ξ . An independent cluster process Φ with center process Π_ξ and component processes $\{(Z^\infty)_x, x \in X\}$, is a self-similar point process.

Example. Branching-stable point processes defined in Zanella & Zuyev (2015).

Multifractal point processes – extension of previous. . .

Thank you!

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